

Rishabh Tennankore, Ph.D.

[Portfolio](#) | [Google Scholar](#) | [Email](#)

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Boulder, CO

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PROFESSIONAL PROFILE

Materials Science Ph.D. with expertise in nanoscale thin-film fabrication and surface engineering for functional device applications. Experienced in multivariable process development using structured DOE (> 500 runs) to correlate process conditions with film properties. Hands-on background in cleanroom fabrication, photolithography, thin-film deposition, and surface metrology (SEM, Profilometry). Data-driven engineer skilled at extracting mechanistic insight from complex datasets to guide process optimization.

SKILLS

Process Engineering & Optimization

- Design of Experiments (DOE), Root-Cause Analysis (RCA), Parameter Optimization

Thin-films and Semiconductors

- *Cleanroom Fabrication & Processing*
 - PVD (e-beam evaporation, sputtering), CVD (Vapor-phase silane surface functionalization; mTorr regime), UV-Ozone, Plasma Bonding, Photolithography (Coating, Photopatterning, Lift-off)
- *Metrology & Characterization*
 - SEM, Profilometry, X-Ray Scattering, Potentiostat, Optical Microscopy, Rheology

Modeling & Data Analysis

- Python, MATLAB, JMP, LabVIEW, MS Excel, AutoCAD, ImageJ, Arduino

Polymer Synthesis

- Free radical and living radical polymerization, Step-growth polymerization, Photopolymerization, Sol-Gel processing, Silicone hydrosilylation

WORK EXPERIENCE

Membrane Development Scientist

Oct 2025 – Dec 2025

OsmoPure Technologies Inc. – Boulder, CO

Led development and scale-up of thin-film coating processes for asymmetric membranes enabling pressure-driven, vapor-transport mediated water purification; owned end-to-end formulation, process optimization, and troubleshooting in a fast-paced startup manufacturing environment.

Highlights:

- **Led >100 TEOS-PDMS sol-gel formulation and process optimization iterations** across multiple substrate suppliers, narrowing to top-performing chemistries and establishing robust processing windows
- **Re-engineered formulations** to reduce reliance on multi-step vapor deposition, **reducing cycle time by > 2 hours** and increasing daily throughput from 3 to > 10 samples, accelerating feedback loops and process iteration
- **Reduced areal thickness variance from > 20% to < 3%** via structured DOE and implementation of standardized profilometry workflows

- **Led RCA on coating cracking** and delamination through SEM and performance testing; implemented corrective controls **eliminating > 90% crazing failures**
- Increased operational pressure tolerance > 2x, improved separation efficiency by 75%, and enhanced permeation rates 10× through three structured optimization campaigns

Graduate Research Assistant

Aug 2020 – Aug 2025

University of Colorado, Materials Science and Engineering – Boulder, CO

Developed stimuli-responsive thin-film polymer systems and microfabricated MEMS prototypes integrating polymer synthesis, materials characterization, fluid dynamics, process modeling and performance validation.

Highlights:

- **Fabricated SU-8 micropillar arrays on silicon wafers via photolithography** and deposited ~ **10 nm Au films via e-beam evaporation and sputtering**, optimizing deposition technique to achieve conformal coatings and pattern fidelity on structured substrates, enabling voltage-tunable surface energy modulation
- **Deposited ~ 100 nm anionic hydrogel coatings on porous substrates**, achieving enhanced ion transport properties; characterized film morphology and electrical properties **using SEM & Resistivity measurements** to establish structure–property relationships
- Designed and integrated tri-layer polymer–substrate stacks using **plasma bonding and solution-processing** to ensure robust interfacial adhesion and defect-free sealing
- Designed and executed multi-variable DOE analysis (**> 500 experimental runs**), defining repeatable operating windows and validating performance hypotheses
- Built MATLAB-analysis pipelines to clean, analyze, and model large experimental datasets to correlate fabrication parameters with device performance

R&D Engineer

Jan 2018 – Dec 2018

BASF – Wyandotte, MI

Highlights:

- **Led cross-functional effort to design and validate methods** to test shear-driven performance across 4 surfactant formulations, established structure–property–performance relationships – informing downstream product development decisions
- Applied CFD modeling, statistical analysis and machine learning methods to **develop a regression model** validating parameter–response correlations and **achieved adjusted R² up to 0.97**

EDUCATION

Doctor of Philosophy (Ph.D.), Materials Science and Engineering

Aug 2020 – Aug 2025

- University of Colorado Boulder; GPA: 3.96 / 4.00

Master of Science (M.S.), Macromolecular Science and Engineering

Sep 2017 – Dec 2018

- University of Michigan, Ann Arbor; GPA: 3.96 / 4.00

PUBLICATIONS

4 peer-reviewed journal articles; additional conference, thesis and submitted work available upon request